Hierarchical modeling for tumor microenvironment signature interactions

Chengxin Yu[1, 2], Kieran R. Campbell[1, 2, 3, 4, 5]

[1] Department of Molecular Genetics, University of Toronto; Canada
[2] Lunenfeld-Tanenbaum Research Institute, Sinai Health; Canada
[3] Department of Statistics, University of Toronto; Canada
[4] Ontario Institute for Cancer Research; Canada
[5] Vector Institute; Canada

Introduction

Pancreatic adenocarcinoma (PDAC) is associated with aggressive tumor and metastasis with an abysmal 5-year survival rate.[1] As a major component of the microenvironment, immune cells can promote disease progression and immunnosuppression in PDAC.[2,3] Different cell lineages constitute a large and heterogeneous population of the immune compartment in the TME. An example is the myeloid cells which have the tumor associated macrophage (TAM) and the myeloid derived suppressive cell (MDSC) subpopulations.[4] Here we seek to identify the immunosuppressive subpopulations via their correlations with phenotypes such as cytotoxic T cell abundance. Here we present a tool under development for systematic investigation into the intraTME interactions.

Project aims

1. Provide an automatic pipeline for extraction of novel transcriptional substates usng LIGER.
2. Develop a Bayesian model to elucidate correlations between cytotoxic T cell abundance and cellular substates.
3. Characterization of immunosuppressive TME subpopulations.

Identification of myeloid subpopulations in PDAC

Our pipeline integrated the LIGER package for transcriptional signatures extraction from a multi-cohort PDAC dataset[5–9] (Fig.1). Signature activation profiles would provide characterization of cell substates (Fig.2).

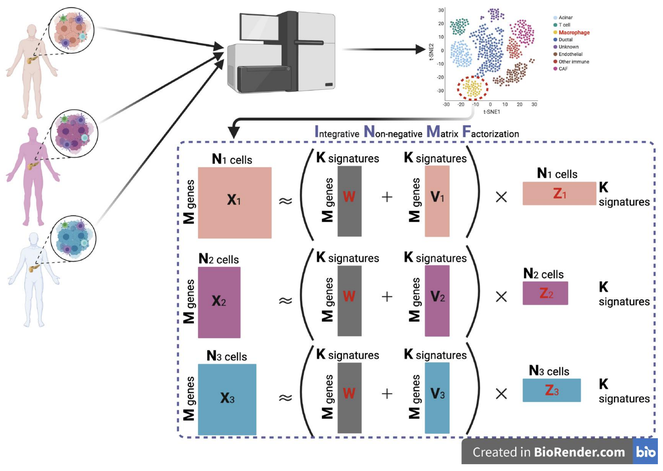

Figure 1: Myeloid signature extraction with LIGER

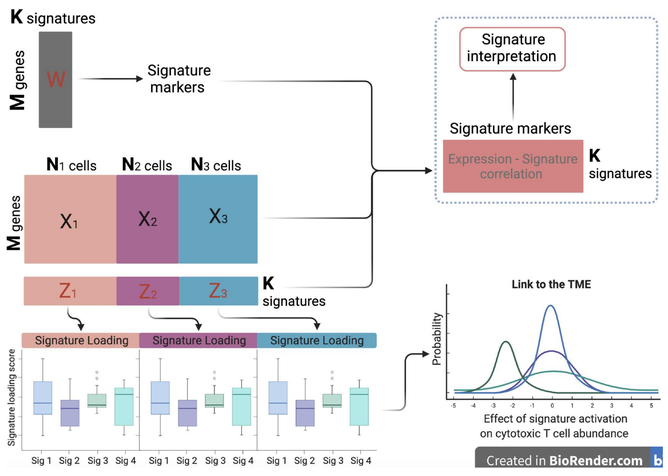

Figure 2: Characterization workflow

Investigation of transcriptional signature activation and marker activation in the tumor microenvironment. We identified 10 subpopulations of myeloid cells in PDAC of which 8 validated between discovery and validation (Fig.3).

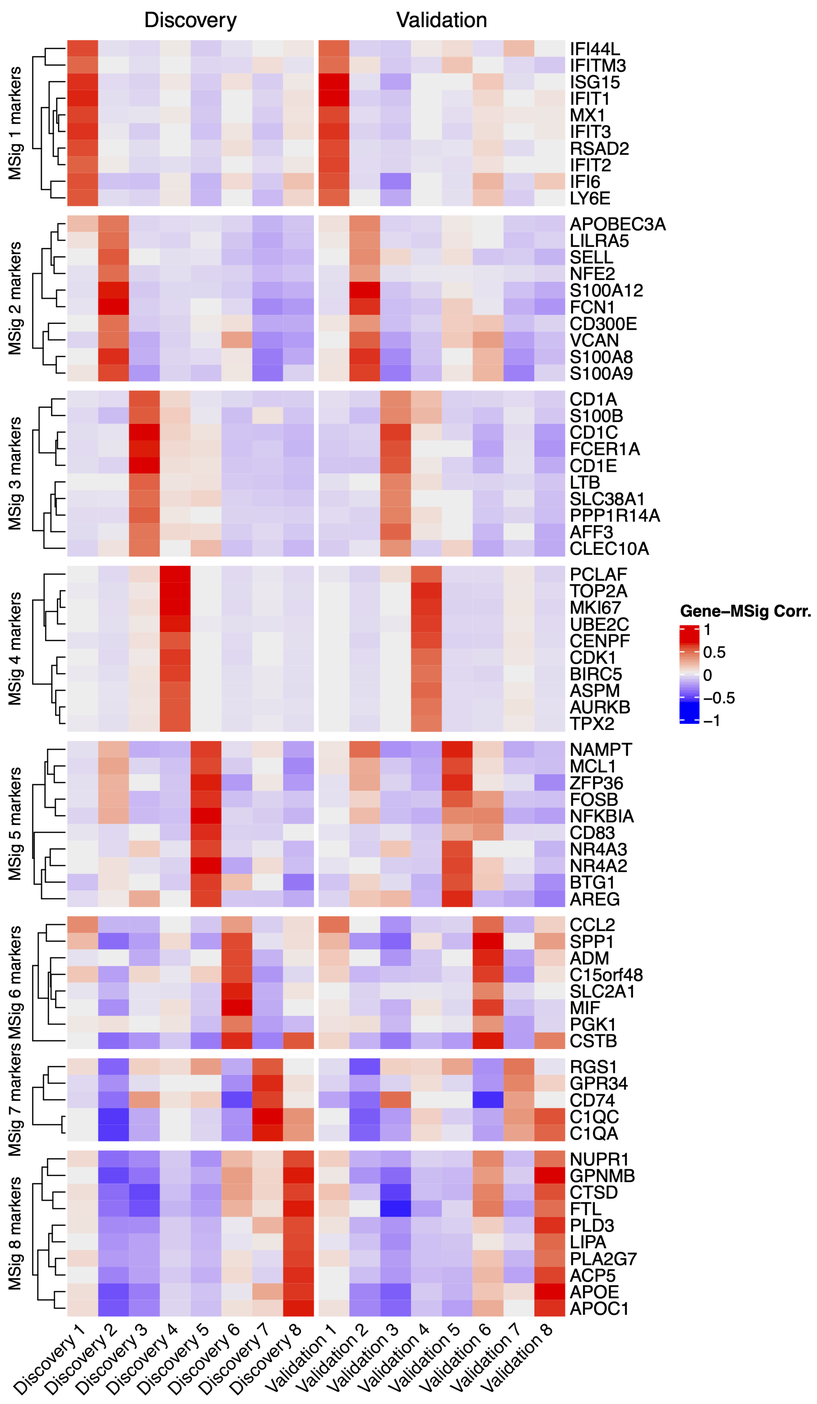

Figure 3: Marker identification for each transcriptional signature with cross-cohort validation. Here we show the correlation between gene activation and signature activation in single-cells. The left column shows validated signatures from discovery cohort, the right column shows validated signatures from validation cohort

A Bayesian model to identify immunosuppressive subpopulations

Single cell level signature weightings provide insights into both cell identity and activated functional programs. Combined with clinical information and Bayesian statistical inference, we used STAN to build a extendable model in our pipeline that links quantitative traits of the tumor microenvironment to clinically relevant phenotypes (Fig.4). Preliminary results confirmed the model's capability to reveal the immunosuppressive TAM and MDSC subpopulations characterized by canonical markers (Fig.5).

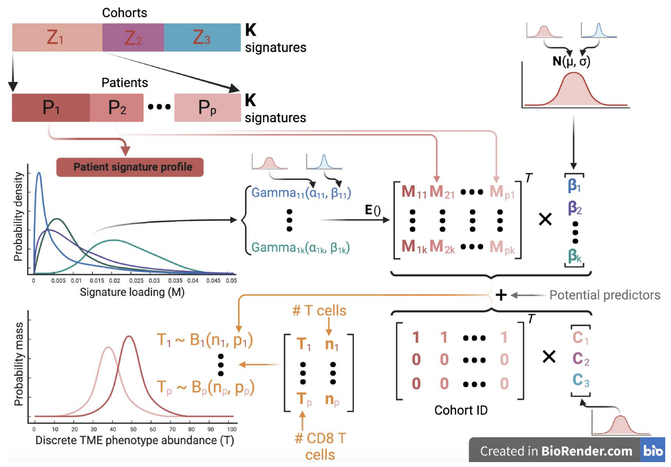

Figure 4: Hierarchical inference model

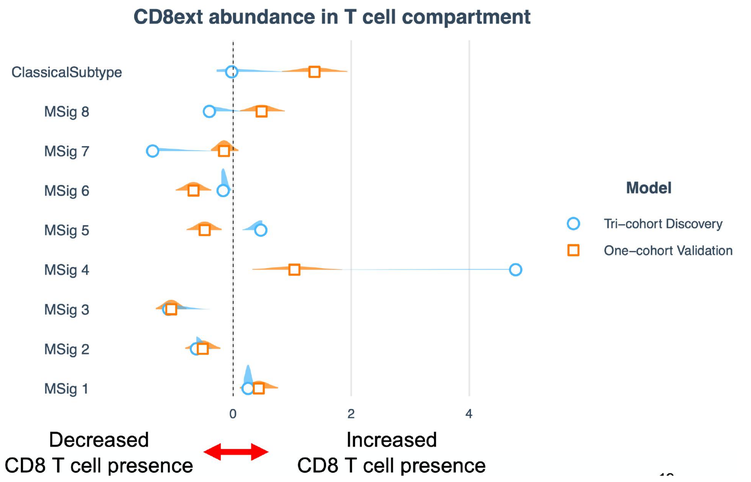

Figure 5: Qualitative inference of the effect of functional and cellular signature activation on TME composition, with control for inter-patient technical variability

Next Steps

- Validation of novel transcriptional signatures and cell substates in PDAC.
- Implementation of reporting and visualization functions.
- Adding support for mouse single-cell RNA sequencing data.

References

1. Kleeff, J. *et al.* Pancreatic cancer. *Nature Reviews Disease Primers* **2**, 16022 (2016).
2. Evans, A. & Costello, E. The role of inflammatory cells in fostering pancreatic cancer cell growth and invasion. *Front. Physiol.* **3**, 270 (2012).
3. Ho, W. J., Jaffee, E. M. & Zheng, L. The tumour microenvironment in pancreatic cancer — clinical challenges and opportunities. *Nat. Rev. Clin. Oncol.* **17**, 527–540 (2020).
4. Zhu, Y. *et al.* Tissue-Resident macrophages in pancreatic ductal adenocarcinoma originate from embryonic hematopoiesis and promote tumor progression. *Immunity* **47**, 323–338.e6 (2017).
5. Welch, J. D. *et al.* Single-Cell multi-omic integration compares and contrasts features of brain cell identity. *Cell* **177**, 1873–1887.e17 (2019).
6. Peng, J. *et al.* Author correction: Single-cell RNA-seq highlights intra-tumoral heterogeneity and malignant progression in pancreatic ductal adenocarcinoma. *Cell Res.* **29**, 777 (2019).
7. Chan-Seng-Yue, M. *et al.* Transcription phenotypes of pancreatic cancer are driven by genomic events during tumor evolution. *Nat. Genet.* **52**, 231–240 (2020).
8. Lin, W. *et al.* Single-cell transcriptome analysis of tumor and stromal compartments of pancreatic ductal adenocarcinoma primary tumors and metastatic lesions. *Genome Med.* **12**, 80 (2020).
9. Steele, N. G. *et al.* Multimodal mapping of the tumor and peripheral blood immune landscape in human pancreatic cancer. *Nature Cancer* **1**, 1097–1112 (2020).